

RBE SSC CGL Tier-1 Live Mock 29, August 31, 2026

Q. 1	What should come in place of the question mark (?) in the given series based on the English alphabetical order? Q GK, M CG, I YC, E UY, A QU, ? a) X MQ  b) W MQ  c) W NP  d) X NP 
Explanation:	Logic: Each position follows −4 (cyclic) pattern. - First letters: Q → M → I → E → A → W - Second letters: G → C → Y → U → Q → M - Third letters: K → G → C → Y → U → Q  Final Answer: WMQ
Q. 2	Which of the following number will replace the question mark (?) in the given series? 3, 8, 24, 48, 120, ? a) 169  b) 143  c) 225  d) 168 
Explanation:	Logic: Next number = square of the next prime − 1 - The series follows: 3, 8, 24, 48, 120, ? - Consecutive primes: 2, 3, 5, 7, 11, 13 - Apply the rule for the next prime (13): $13^2 - 1 = 169 - 1 = 168$ Answer: 168

Q. 3

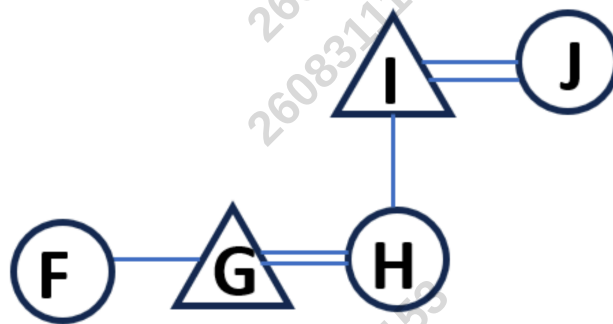
F is the sister of G. G is the husband of H. H is the daughter of I. I is the husband of J. How is F related to J?

a) Daughter's husband's sister ☒

b) Husband's sister ☐

c) Daughter's husband's mother ☐

d) Husband's mother ☐



Explanation:

Family relations breakdown:

- F is the sister of G
- G is the husband of H
- H is the daughter of I
- I is the husband of J

So:

- F → sister of G → G's wife = H → H's mother = J
- F is **sister of H's husband**

☒ Correct answer: **Daughter's husband's sister**

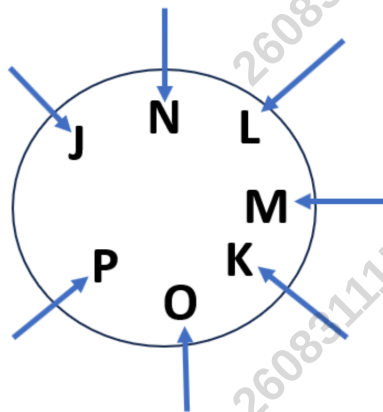
Q. 4	JK 17 is related to GH 6 in a certain way. In the same way, MP 12 is related to JM 1. To which of the given options is DO –6 related, following the same logic? a) LA –21  b) AL –17  c) GA –21  d) BN–21 
Explanation:	Logic : Letter shift: Each letter moves 3 positions backward in the alphabet. - JK → GH, MP → JM → DO → AL Number shift: Subtract 11 from the given number. 17 → 6, 12 → 1 → –6 → –17 Answer: AL –17
Q. 5	In a certain code language, 'OWNS' is coded as '6391' and 'SIDE' is coded as '2468'. What is the code for 'S' in the given code language? a) 3  b) 9  c) 6  d) 2 
Explanation:	OWNS → 6391 SIDE → 2468 The common letter in both words is S. The common number that appears in the code of both words is 6.  Therefore, by matching the common letter with the common number, the code for S = 6.

Q. 6

J, K, L, M, N, O and P are sitting around a circular table facing the centre. K is sitting to the immediate right of O and immediate left of M. L is sitting to the immediate right of M. N is sitting to the immediate left of J. P is sitting to the immediate right of J and immediate left of O. Who is second left of L.

- a) O ☐
- b) M ☐
- c) P ☐
- d) K ☒

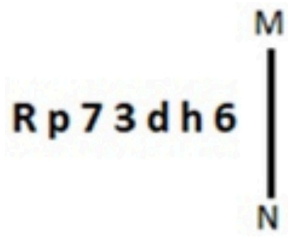
Explanation:



Clearly K is second left of L.

Q. 7

Select the correct mirror image of the given figure when the mirror is placed at MN.



a) Rp73dh6



b) Rb73qhe



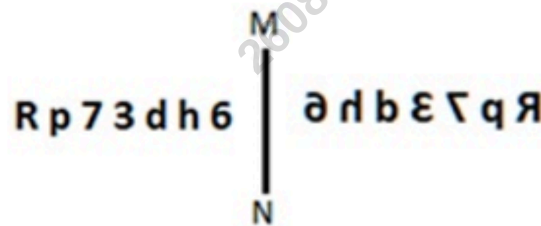
c) Rb73qhe



d) Rp73qhe



Explanation:



Q. 8

Muskan starts from Point A and drives 12 km towards East. He then takes a right turn, drives 8 km, turns right and drives 14 km. He then takes a right turn and drives 11 km. He takes a final right turn, drives 2 km and stops at Point P. How far (shortest distance) and towards which direction should he drive in order to reach Point A again? (All turns are 90° turns only, unless specified.)

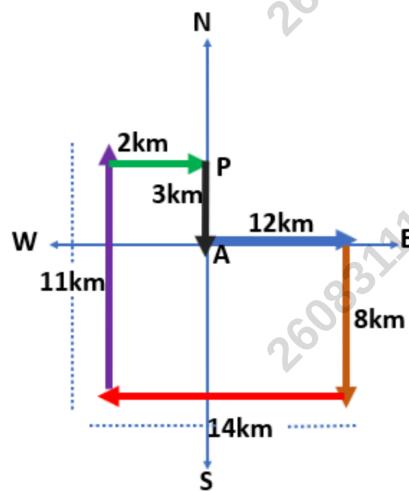
a) 3 km to the south ✓

b) 1 km to the south ✗

c) 4 km to the south ✗

d) 2 km to the south ✗

Explanation:



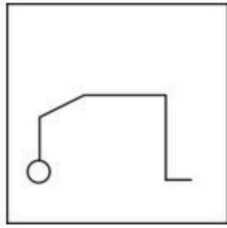
Muskan has to walk 3 km south to reach point A again from point P.

Q. 9	FV 14 is related to JZ 7 in a certain way. In the same way, DZ 18 is related to HD 11. To which of the given options is SU 4 related, following the same logic? a) WY -3  b) XW -4  c) XY -3  d) WY -4 
Explanation:	Letters shift +4 $F \rightarrow J, V \rightarrow Z$ $D \rightarrow H, Z \rightarrow D$ Number rule = -7 $14 \rightarrow 7$ $18 \rightarrow 11$ Apply to SU 4 $S \rightarrow W (+4)$ $U \rightarrow Y (+4)$ $4 - 7 = -3$  Final Answer: WY -3
Q. 10	Kaziranga National Park : Assam :: Gir National Park : ? a) Rajasthan  b) Gujarat  c) Madhya Pradesh  d) Uttarakhand 
Explanation:	Kaziranga National Park is located in Assam. Similarly, Gir National Park is located in Gujarat. Final Answer: 2) Gujarat

Q. 11	In a certain code language, "SMELL" is coded as "86" and "SARAJ" is coded as "88". What is the code for "KRISH" in that language? a) 65  b) 75  c) 90  d) 102 
Explanation:	Logic: Code = Sum of letter positions + (Number of letters)² • SMELL $\rightarrow 19+13+5+12+12 = 61$; $5^2 = 25 \rightarrow 61+25 = 86$ • SARAJ $\rightarrow 19+1+18+15+10 = 63$; $5^2 = 25 \rightarrow 63+25 = 88$ • KRISH $\rightarrow 11+18+9+19+8 = 65$; $5^2 = 25 \rightarrow 65+25 = 90$ Answer: 90

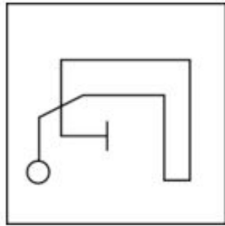
Q. 12

Select the option figure in which the given figure (X) is embedded (rotation is NOT allowed).

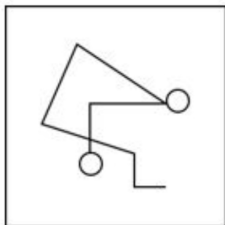


(X)

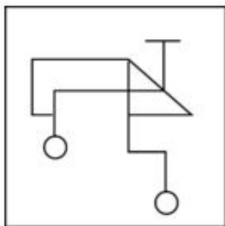
a)



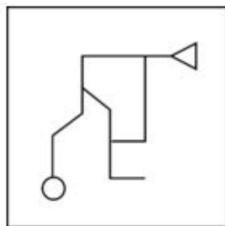
b)



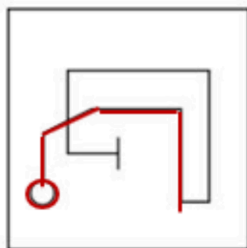
c)



d)



Explanation:



Q. 13

Select the triad that follows the same pattern as that followed by the two triads given below. Both triads follow the same pattern.

BB-CD-EF

KL-MN-QP

a) CD-EF-FH



b) EF-FG-GH



c) QR-ST-WV



d) IJ-KL-MN

**Explanation:****Logic:**

- First letters → Prime positions
- Second letters → Consecutive even positions

1. CD-EF-FH  (F not prime)2. EF-FG-GH  (F not prime, G not even)3. QR-ST-WV  (17,19,23 prime & 18,20,22 even)4. IJ-KL-MN  (I not prime) **Final Answer: QR – ST – WV**

Q. 14	This question is based on the five three-digit numbers given below (Left): 928, 564, 731, 482, 859 (Right) Example – 731 → First digit = 7, second digit = 3 and third digit = 1 (NOTE: All operations are to be done from left to right.) What will be the result if the second digit of the highest number is subtracted from the third digit of the lowest number? a) 1  b) 0  c) -1  d) 2 
Explanation:	Highest = 928 Lowest = 482 Second digit of highest = 2 Third digit of lowest = 2 $2 - 2 = 0$ Final Answer: 0

Q. 15

If 1 is added to each even digit and 3 is subtracted from each odd digit in the number 58372649, how many digits will appear more than once in the new number thus formed?

- a) 0 
- b) 1 
- c) 2 
- d) 3 

Given number: 58372649

Rule:

- +1 to each **even** digit
- -3 from each **odd** digit

$$5 - 3 = 2$$

$$8 + 1 = 9$$

$$3 - 3 = 0$$

$$7 - 3 = 4$$

$$2 + 1 = 3$$

$$6 + 1 = 7$$

$$4 + 1 = 5$$

$$9 - 3 = 6$$

New number formed:

2 9 0 4 3 7 5 6

All digits are different.

Final Answer: Zero

Explanation:

Q. 16	In a certain code language, 'how are you' is coded as 'pk rd rc' and 'are they coming' is coded as 'rd qn bt'. How is 'are' coded in the given language? a) rd  b) qn  c) pk  d) bt 
Explanation:	Logic Summary: - Identify the common word in both phrases: 'are' - Find the common code in both coded phrases: rd - Match them: 'are' → rd Answer: rd
Q. 17	67 is related to 80 following a certain logic. Following the same logic, 13 is related to 17. To which of the following is 52 related, following the same logic? a) 75  b) 56  c) 59  d) 61 
Explanation:	Logic : - Rule: Next number = Number + Sum of its digits $67 \rightarrow 6 + 7 = 13 \rightarrow 67 + 13 = 80$ $13 \rightarrow 1 + 3 = 4 \rightarrow 13 + 4 = 17$ $52 \rightarrow 5 + 2 = 7 \rightarrow 52 + 7 = 59$ Answer: 59

Q. 18

Which of the following number will replace the question mark (?) in the given series?

156, 107, 43, -38, -138, ?

a) -259 

b) -214 

c) -236 

d) -246 

Explanation:

Series: 156, 107, 43, -38, -138, ?

Logic: Subtract consecutive perfect squares

- $156 - 7^2 = 156 - 49 = 107$
- $107 - 8^2 = 107 - 64 = 43$
- $43 - 9^2 = 43 - 81 = -38$
- $-38 - 10^2 = -38 - 100 = -138$
- $-138 - 11^2 = -138 - 121 = -259$

Answer: -259

Q. 19

Based on the English alphabetical order, three of the following four letter-clusters are alike in a certain way and thus form a group. Which letter-cluster DOES NOT belong to that group? (Note: The odd one out is not based on the number of consonants/vowels or their position in the letter-cluster.)

- a) GX-YF 
- b) LS-ZD 
- c) JU-WH 
- d) NQ-VI 

Explanation:

- Take the **alphabet positions** of the letters: A=1, B=2, ... Z=26.
- **Sum of first two letters = Sum of last two letters**

Check each cluster:

1. GX-YF

- $G + X = 7 + 24 = 31$
- $Y + F = 25 + 6 = 31$ ✓

2. LS-ZD

- $L + S = 12 + 19 = 31$
- $Z + D = 26 + 4 = 30$ 

3. JU-WH

- $J + U = 10 + 21 = 31$
- $W + H = 23 + 8 = 31$ ✓

4. NQ-VI

- $N + Q = 14 + 17 = 31$
- $V + I = 22 + 9 = 31$ ✓

 Only LS-ZD fails the rule.

Answer: LS-ZD

Q. 20

Based on the English alphabetical order, three of the following four letter-cluster pairs are alike in a certain way and thus form a group. Which letter-cluster pair DOES NOT belong to that group? (Note: The odd one out is not based on the number of consonants/vowels or their position in the letter cluster.)

a) RY-NU ✗

b) FM-BI ✗

c) CV-YS ✓

d) LQ-HM ✗

Explanation:

Check options:

1. RY → NU

• $R(18) - 4 = N(14)$ ✓• $Y(25) - 4 = U(21)$ ✓

✓ Follows logic

2. FM → BI

• $F(6) - 4 = B(2)$ ✓• $M(13) - 4 = I(9)$ ✓

✓ Follows logic

3. CV → YS

• $C \rightarrow Y$ (-4 cyclically), $V \rightarrow R$ (-4), so CV should become YR, not YS. Hence Option 3 is the odd one out. ✗

4. LQ → HM

• $L(12) - 4 = H(8)$ ✓• $Q(17) - 4 = M(13)$ ✓

✓ Follows logic

Answer: CV → YS

Q. 21

If + means −, − means ×, × means ÷, and ÷ means +, then what will come in place of the question mark (?) in the following equation?

$$47 + 5 \div 40 \times 10 - 2 = ?$$

a) 50 b) 52 c) 55 d) 49 **Explanation:****Logic :**

Replace symbols according to the rule:

 $\rightarrow -, - \rightarrow \times, \times \rightarrow \div, \div \rightarrow +$ Original equation: $47 + 5 \div 40 \times 10 - 2$ After replacement: $47 - 5 + 40 \div 10 \times 2$

Solve step by step:

$$40 \div 10 = 4 \rightarrow 4 \times 2 = 8$$

$$47 - 5 = 42 \rightarrow 42 + 8 = 50$$

Answer: 50

Q. 22

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusion(s) logically follow(s) from the statements.

Statements:

All birds are fruits.

No fishes are fruits.

All fishes are toys.

Conclusions:

(I) Some toys are not birds.

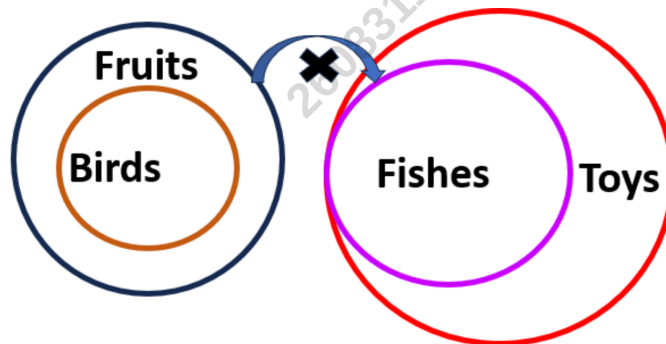
(II) No fruits are toys.

a) Both conclusions (I) and (II) follow. ✗

b) Only conclusion (II) follows. ✗

c) Neither conclusion (I) nor (II) follows. ✗

d) Only conclusion (I) follows. ✓



Explanation:

(I) Some toys are not birds

All fishes are toys.

No fishes are fruits.

All birds are fruits.

So fishes (toys) cannot be birds.

Hence, some toys (fishes) are not birds. ✓

Conclusion (I) follows.

(II) No fruits are toys

We only know fishes (toys) are not fruits.

But it is not given that all toys are fishes.

So fruits may or may not be toys.

✗ **Conclusion (II) does not follow.**

Q. 23

In a class of 80 students, A is ranked 31st from the left end and B is ranked 27th from the right end. A and B exchange their seats. After the exchange, what will be the rank of A from the left end?

- a) 52nd 
- b) 53rd 
- c) 54th 
- d) 55th 

Explanation:

Total students = 80

A is 31st from left

B is 27th from right

Convert B's position into left position:

Left position of B =

$$80 - 27 + 1$$

$$= 54\text{th from left}$$

After exchange, A sits at B's original position.

So A's new rank from left = 54th

Final Answer: 54th

Q. 24

What should come in place of the question mark (?) in the given series based on the English alphabetical order?

ABC, BEG, CHK, DKO, ENS, ?

- a) FRV 
- b) FQW 
- c) GRW 
- d) GSV 

Explanation:

Series: ABC, BEG, CHK, DKO, ENS, ?

Logic:

- 1st letters: +1 → A, B, C, D, E → next **F**
- 2nd letters: +3 → B, E, H, K, N → next **Q**
- 3rd letters: +4 → C, G, K, O, S → next **W**

Next term = FQW

Q. 25

Six people, G, H, I, J, K, and L, live on six different floors of the same building. The lowermost floor in the building is numbered 1, the floor above it, number 2, and so on till the topmost floor is numbered 6. L lives on an even-numbered floor but not 6. The sum of floors on which J and L live is 7. H lives immediately above G. The sum of floors on which L and I live is 3. How many people live between H and L?

a) 1



b) 0



c) 2



d) 3



Explanation:

6 TH FLOOR	K
5 TH FLOOR	J
4 TH FLOOR	H
3 RD FLOOR	G
2 ND FLOOR	L
1 ST FLOOR	I

Only one person between H & L.

Q. 26	Read the statements regarding the Indian Monsoon mechanism: 1. The El Niño phenomenon generally weakens the south-west monsoon rainfall in India. 2. The Indian Ocean Dipole (IOD) in its positive phase enhances monsoon precipitation. Which of the statements given above is/are correct? a) 1 only  b) 2 only  c) Both 1 & 2  d) Neither 1 & 2 
Explanation:	 Correct Answer: Option C — Both 1 & 2 Explanation: 1. El Niño phenomenon → ◦ El Niño refers to abnormal warming of the central/eastern Pacific Ocean. ◦ It weakens the southwest monsoon rainfall in India by disrupting the Walker circulation and reducing moisture transport. ◦ Historically, many drought years in India (e.g., 2002, 2009, 2015) coincided with El Niño events. ◦  Statement 1 is correct. 2. Indian Ocean Dipole (IOD) → ◦ Positive IOD = Warmer western Indian Ocean & cooler eastern Indian Ocean near Indonesia. ◦ This enhances monsoon precipitation over India by strengthening westerly winds and moisture inflow. ◦ Even during El Niño years, a strong positive IOD can offset rainfall deficiency (e.g., 1997). ◦  Statement 2 is correct.

Q. 27	Which of the following statements regarding the Mughal agrarian system under Akbar is INCORRECT? a) Polaj was land cultivated annually without being left fallow for any single cropping season. ✗ b) Parauti was land left uncultivated temporarily for a short duration to recover its natural fertility. ✗ c) Chachar was fertile land that was assessed at the highest tax rate under the Dahsala system. ✓ d) Banjar was wasteland left uncultivated for five or more consecutive years. ✗
Explanation:	✓ Correct Answer: Option C — Chachar was fertile land that was assessed at the highest tax rate under the Dahsala system. Explanation: Under Akbar's Dahsala system (introduced by Raja Todar Mal in 1580 CE), land was classified into four categories for revenue assessment: Polaj ✓ - Land cultivated annually without being left fallow. - Considered the most fertile and productive. Parauti (Parati) ✓ - Land left uncultivated temporarily for 1–2 years to regain fertility. Chachar ✗ (Incorrect statement in the question) - Actually refers to land left fallow for 3–4 years. - It was not fertile land taxed at the highest rate. - Revenue demand was lower compared to Polaj. Banjar ✓ - Wasteland left uncultivated for 5 years or more. - Generally not assessed for revenue until reclaimed.

Q. 28	Which of the following human brain regions is INCORRECTLY matched with its primary function? a) Cerebrum — Center for intelligence, memory, reasoning, and voluntary muscle control ✗ b) Cerebellum — Regulation of posture, bodily balance, and precision of voluntary movements ✗ c) Hypothalamus — Control of body temperature, urge for eating/drinking, and hormone secretion ✗ d) Medulla Oblongata — Primary control center for conscious logical thought and decision making ✓
Explanation:	✓ Correct Answer: Option D — Medulla Oblongata: Primary control center for conscious logical thought and decision making. Explanation: Cerebrum ✓ - Largest part of the brain. - Functions: intelligence, memory, reasoning, voluntary muscle control, emotions, and conscious thought. - Correctly matched. Cerebellum ✓ - Located at the back of the brain. - Functions: regulation of posture, balance, coordination, and precision of voluntary movements. - Correctly matched. Hypothalamus ✓ - Small but vital region below the thalamus. - Functions: control of body temperature, hunger, thirst, circadian rhythms, and regulation of hormone secretion via pituitary gland. - Correctly matched. Medulla Oblongata ✗ - Located in the brainstem. - Functions: involuntary control of vital processes such as heartbeat, breathing, blood pressure, swallowing, and reflexes. - The statement in the question is incorrect because conscious logical thought and decision making are functions of the cerebrum, not the medulla.

Q. 29	Which of the following statements best describes a Vostro Account? a) An account held by a domestic bank in a foreign bank in foreign currency. b) An account held by a domestic bank in another domestic bank in domestic currency. c) An account held by a foreign bank in a domestic bank in domestic currency. d) An account held by an individual in a foreign bank to manage personal investments.
Explanation:	 Correct Answer: Option C — A Vostro Account is an account held by a foreign bank in a domestic bank in domestic currency. Explanation: • Vostro Account → ◦ "Vostro" means "yours" in Latin. ◦ It refers to an account that a foreign bank holds in a domestic bank, denominated in the domestic currency of the country where the domestic bank is located. ◦ Example: A US bank maintains a Vostro Account with an Indian bank in Indian Rupees (INR). • Nostro Account (for comparison): ◦ "Nostro" means "ours." ◦ It is an account that a domestic bank holds in a foreign bank, denominated in the foreign currency. ◦ Example: An Indian bank maintains a Nostro Account with a US bank in US Dollars (USD).

Q. 30	What type of orbit is utilized by India's NavIC navigational satellite system? a) Low Earth Orbit  b) Geosynchronous & inclined geosynchronous Orbit  c) Polar Sun-Synchronous Orbit  d) Medium Earth Orbit 
Explanation:	India's <u>NavIC</u> (https://www.isro.gov.in/SatelliteNavigationServices.html) (Navigation with Indian Constellation) system uses Geosynchronous & inclined geosynchronous Orbit (along with Geostationary orbits). Out of the seven operational satellites in its constellation, three are placed in Geostationary Earth Orbit (GEO) and four in Inclined Geosynchronous Orbit (IGSO). Orbital Details • Geostationary Satellites (GEO): Three satellites stay fixed over specific points on the equator. • Inclined Geosynchronous Satellites (IGSO): Four satellites move in figure-eight type paths at an inclination angle of 29 degrees to provide continuous regional coverage over India and surrounding areas.

Q. 31	Arrange the following 19th-century tribal and peasant uprisings in correct chronological order: 1. Santhal Rebellion 2. Indigo Revolt 3. Deccan Riots 4. Munda Ulgulan a) 3 — 1 — 2 — 4  b) 1 — 3 — 2 — 4  c) 2 — 1 — 4 — 3  d) 1 — 2 — 3 — 4 
Explanation:	 Correct Answer: Option D — 1, 2, 3, 4 Explanation: 1. Santhal Rebellion (1855–56) ◦ Tribal uprising in present-day Jharkhand against British colonial rule and exploitation by moneylenders and zamindars. 2. Indigo Revolt (1859–60) ◦ Peasant movement in Bengal against oppressive indigo planters who forced peasants to grow indigo instead of food crops. 3. Deccan Riots (1875) ◦ Peasant uprising in Maharashtra against moneylenders due to debt bondage and exploitation. 4. Munda Ulgulan (1899–1900) ◦ Led by Birsa Munda in Chotanagpur region, Jharkhand. ◦ Revolt against British policies, missionary influence, and exploitation of tribal land rights.

Q. 32

Match the Ore in List-I with its primary metal content in List-II:

Ore	Metal
A. Bauxite	1. Lead
B. Galena	2. Aluminum
C. Cinnabar	3. Copper
D. Malachite	4. Mercury

a) A-3, B-1, C-4, D-2

b) A-1, B-2, C-3, D-4

c) A-2, B-1, C-4, D-3

d) A-2, B-4, C-1, D-3

☒ **Correct Answer:** Option C — A-2, B-1, C-4, D-3

Explanation:

Let's match each ore with its primary metal content:

Explanation:

- **Bauxite** → Chief ore of **Aluminum (Al)**.
- **Galena** → Chief ore of **Lead (Pb)**.
- **Cinnabar** → Chief ore of **Mercury (Hg)**.
- **Malachite** → Important ore of **Copper (Cu)**.

Q. 33	Why do ionic compounds generally exhibit higher melting points than covalent molecular compounds? a) Ionic substances contain continuous hydrogen bonds that weaken rapidly when heated slightly.  b) Electrostatic forces of attraction between oppositely charged ions require vast energy to break.  c) Covalent compounds contain permanent metallic bonds that break easily at low thermal states.  d) Ionic bonds involve sharing valence electrons across extensive three-dimensional lattice planes. 
Explanation:	 Correct Answer: Option B — Electrostatic forces of attraction between oppositely charged ions require vast energy to break. Explanation: • Ionic Compounds ◦ Formed by strong electrostatic forces of attraction between cations and anions. ◦ These forces extend across a three-dimensional crystalline lattice, making them very strong and stable. ◦ Breaking these bonds requires large amounts of energy, hence ionic compounds generally have high melting and boiling points. • Covalent Compounds ◦ Formed by sharing of electrons between atoms. ◦ While covalent bonds inside molecules are strong, the intermolecular forces (like van der Waals or hydrogen bonds) are relatively weak. ◦ This leads to lower melting and boiling points compared to ionic compounds.

Q. 34	How does a "climax community" conceptually differ from earlier "pioneer stages" in ecological succession? a) It exhibits higher species diversity, complex nutrient cycles, and steady-state structural stability. ✓ b) It relies entirely on external nutrient inputs to sustain its high primary biological productivity. ✗ c) It consists predominantly of short-lived species with fast reproduction rates and simple food chains. ✗ d) It experiences rapid biomass fluctuations caused by minor changes in seasonal weather patterns. ✗
Explanation:	✓ Correct Answer: Option A — It exhibits higher species diversity, complex nutrient cycles, and steady-state structural stability. Explanation: • Climax Community ◦ Represents the final stable stage of ecological succession. ◦ Features: high species diversity, complex food webs, balanced nutrient cycles, and relative structural stability. ◦ Energy flow and biomass remain in a dynamic equilibrium. • Pioneer Stages ◦ Early stages of succession. ◦ Dominated by short-lived, fast-reproducing species (e.g., lichens, mosses, grasses). ◦ Simple food chains, rapid biomass fluctuations, and dependence on external nutrient inputs. ◦ Ecosystem is unstable and vulnerable to disturbances.

Q. 35	Which of the following pairs regarding Parliamentary Committees is INCORRECTLY matched? a) Public Accounts Committee — 22 Members (15 Lok Sabha, 7 Rajya Sabha) ✗ b) Committee on Estimates — 30 Members (All drawn exclusively from Lok Sabha) ✗ c) Business Advisory Committee — Chaired ex-officio by the Presiding Officer of the House ✗ d) Committee on Public Undertakings — 22 Members (All drawn exclusively from Rajya Sabha) ✓
Explanation:	✓ Correct Answer: Option D — Committee on Public Undertakings: 22 Members (All drawn exclusively from Rajya Sabha) Explanation: Let's check each committee: Public Accounts Committee (PAC) ✓ - Strength: 22 members (15 from Lok Sabha + 7 from Rajya Sabha). - Correctly matched. Committee on Estimates ✓ - Strength: 30 members, all drawn exclusively from Lok Sabha. - Correctly matched. Business Advisory Committee ✓ - Chaired ex-officio by the Presiding Officer of the House (Speaker in Lok Sabha, Chairman in Rajya Sabha). - Correctly matched. Committee on Public Undertakings ✗ - Strength: 22 members (15 from Lok Sabha + 7 from Rajya Sabha). - The statement "All drawn exclusively from Rajya Sabha" is incorrect.

Q. 36	In India, which financial institution acts as the regulator for the housing finance sector? a) SIDBI  b) NHB  c) SEBI  d) NABARD 
Explanation:	 Correct Answer: Option B — NHB (National Housing Bank) Explanation • National Housing Bank (NHB)  ◦ Established in 1988 under the NHB Act, 1987. ◦ It was set up as the apex financial institution for housing finance in India. ◦ Functions: Regulates housing finance companies (HFCs), provides refinance support, and promotes housing finance institutions. • Recent Change: ◦ In 2019, NHB became a fully owned subsidiary of the Reserve Bank of India (RBI). ◦ However, the regulatory authority for Housing Finance Companies (HFCs) was transferred to the RBI. ◦ Despite this, NHB remains the primary institution historically associated with housing finance regulation. Why Other Options Are Wrong • SIDBI  → Focuses on small industries development, not housing finance. • SEBI  → The Securities and Exchange Board of India (SEBI) works as the chief regulator and watchdog for India's securities and capital markets. • NABARD  → Apex bank for agriculture and rural development, not housing finance.

Q. 37	What happens to the resistance of a pure semiconductor material when its temperature increases? a) It increases linearly  b) It decreases  c) It remains constant  d) It becomes infinite 
Explanation:	 Correct Answer: Option B — It decreases Explanation: • Pure Semiconductors (like silicon or germanium) have relatively few free charge carriers at low temperature. • As temperature increases, more electrons gain sufficient energy to jump from the valence band to the conduction band. • This increases the number of charge carriers (electrons and holes) available for conduction. • Since electrical resistance is inversely proportional to the number of charge carriers, the resistance decreases with rising temperature.

Q. 38	By which Constitutional Amendment Act was the voting age in India reduced from 21 years to 18 years? a) 44th Constitutional Amendment Act ❌ b) 52nd Constitutional Amendment Act ❌ c) 61st Constitutional Amendment Act ✔️ d) 86th Constitutional Amendment Act ❌
Explanation:	✔️ Correct Answer: Option C — 61st Constitutional Amendment Act Explanation • The 61st Constitutional Amendment Act, 1988 lowered the voting age in India from 21 years to 18 years. • It amended Article 326 of the Constitution, which deals with elections to the Lok Sabha and State Legislative Assemblies based on adult suffrage. • The rationale was to increase youth participation in the democratic process and strengthen representative governance. Why Other Options Are Wrong • 44th Amendment (1978) ❌ → Restored Fundamental Rights curtailed during Emergency, but did not change voting age. • 52nd Amendment (1985) ❌ → Introduced the anti-defection law, not related to voting age. • 86th Amendment (2002) ❌ → Made education a Fundamental Right for children aged 6–14, not related to voting age.

Q. 39	Which blood vessel carries oxygenated blood from the lungs back into the left atrium of the human heart? a) Pulmonary Artery  b) Aorta  c) Pulmonary Vein  d) Superior Vena Cava 
Explanation:	The correct answer option is Pulmonary Vein. Why this option is correct: •  Pulmonary Vein: This specialized blood vessel receives oxygen-rich (oxygenated) blood directly from the lungs and delivers it straight to the left atrium of the heart. Although most veins carry oxygen-poor blood, the pulmonary veins are a unique exception because they travel <i>towards</i> the heart while carrying oxygenated blood. Why the other options are incorrect: •  Pulmonary Artery: This vessel carries deoxygenated (oxygen-poor) blood away from the right ventricle of the heart and delivers it into the lungs to receive oxygen. •  Aorta: This is the largest artery in the human body, responsible for carrying oxygenated blood away from the left ventricle to distribute it to the rest of the body's tissues. •  Superior Vena Cava: This large vein collects deoxygenated blood from the upper half of the body (head, neck, and arms) and returns it to the right atrium.

Q. 40	Which ruler of the Slave Dynasty established the military department named Diwan-i-Arz to strengthen the army against Mongol invasions? a) Qutb-ud-din Aibak  b) Ghiyas-ud-din Balban  c) Razia Sultana  d) Shams-ud-din Iltutmish 
Explanation:	Explanation:  Option Ghiyas-ud-din Balban: Ghiyas-ud-din Balban (who ruled from 1266 to 1287 CE) established the centralized military department known as Diwan-i-Arz. He separated it from the financial control of the Wazir (Diwan-i-Wizarat) to create a highly efficient, well-trained, and loyal standing army. This reform was critical for safeguarding the Delhi Sultanate against continuous Mongol incursions along the northwestern frontier and for curbing internal rebellions by regional nobles. Incorrect Options Explanation:  Option Qutb-ud-din Aibak: He was the founder of the Slave (Mamluk) Dynasty in 1206 CE. His brief four-year reign was dedicated primarily to establishing initial control and constructing early monuments like the Quwwat-ul-Islam mosque rather than setting up specialized administrative ministries.  Option Razia Sultana: She was the first and only female sovereign of the Delhi Sultanate (ruling from 1236 to 1240 CE). Her short reign was entirely consumed by heavy political turbulence, internal power struggles, and intense opposition from the elite Turkish nobility group known as Chahalgani.  Option Shams-ud-din Iltutmish: Regarded as the real consolidator of the Delhi Sultanate, his major structural implementations included creating the corporate nobility system of forty loyal slaves (Turkan-i-Chahalgani or Chalisa) and introducing the land revenue Iqta system. He did not create the independent Diwan-i-Arz ministry.

Q. 41	In national income accounting, Gross National Product (GNP) at market prices is calculated by adding what component to Gross Domestic Product (GDP)? a) Net Indirect Taxes  b) Depreciation  c) Personal Disposable Income  d) Net Factor Income from Abroad 
Explanation:	 Correct Answer: Option D — Net Factor Income from Abroad Explanation: • Gross Domestic Product (GDP) → Measures the total value of goods and services produced within the domestic territory of a country. • Gross National Product (GNP) → Extends GDP by including Net Factor Income from Abroad (NFIA). ◦ Formula: GNP at Market Price = GDP at Market Price + Net Factor Income from Abroad • NFIA = Income earned by residents from abroad – Income earned by foreigners within the domestic territory. • Market Price → Already accounts for indirect taxes – subsidies, so the key addition to move from GDP → GNP is NFIA.

Q. 42	The historic Lucknow Pact of 1916 was signed between which two major political groups? a) Indian National Congress and British Government  b) Indian National Congress and Muslim League  c) Moderates and Extremists inside the Congress  d) Swaraj Party and Forward Bloc along with INA 
Explanation:	 Correct Answer: Option B — Indian National Congress and Muslim League Explanation - The Lucknow Pact of 1916 was a landmark agreement between the Indian National Congress (INC) and the All India Muslim League. - Context: - Both groups sought constitutional reforms and greater Indian participation in governance under British rule. - The pact was signed during the joint session at Lucknow in December 1916. Key Features: - Congress and Muslim League agreed to present joint demands to the British government. - The League secured separate electorates for Muslims, while Congress accepted this to ensure Hindu-Muslim unity. - It marked the first time both organizations came together on a common political platform. Why Other Options Are Wrong Option A  → INC did not sign a pact with the British Government; it was with the Muslim League. Option C  → Moderates and Extremists reunited at the Lucknow Session (1916), but that was within Congress, not the Lucknow Pact itself. Option D  → Swaraj Party, Forward Bloc, and INA emerged later (1920s–40s), not in 1916.

Q. 43	The term Chukker is associated with which of the following sports? a) Polo ✓ b) Golf ✗ c) Billiards ✗ d) Squash ✗
Explanation:	✓ Correct Answer: Option A — Polo Explanation: • The term Chukker (sometimes spelled Chukka) is used in the sport of Polo. • It refers to a period of play in a Polo match, similar to an inning or quarter in other sports. • Each chukker lasts about 7 minutes, and a standard Polo match typically consists of 4 to 8 chukkers depending on the level of play. • Horses are usually changed after each chukker to prevent exhaustion. Why Other Options Are Wrong • Golf ✗ → Uses terms like par, birdie, eagle, not chukker. • Billiards ✗ → Uses terms like frames, shots, breaks, not chukker. • Squash ✗ → Uses games, rallies, points, not chukker.

Q. 44

Which of the following statements regarding electromagnetic principles is INCORRECT?

- a) Lenz's Law dictates that induced electric current flows to oppose the change in flux causing it. ✗
- b) Faraday's Law states that induced electromotive force equals the rate of change of magnetic flux. ✗
- c) Lenz's Law serves as a direct practical application of the law of conservation of momentum. ✓
- d) Transformer devices operate on mutual electromagnetic induction to alter alternating voltage levels. ✗

✓ **Correct Answer:** Option C — Lenz's Law serves as a direct practical application of the law of conservation of momentum.

Explanation:

Explanation:

1. **Lenz's Law** ✓
 - States that the induced current always flows in such a direction that it **opposes the change in magnetic flux** producing it.
 - This is correct and is a direct consequence of the **law of conservation of energy**, not momentum.
2. **Faraday's Law** ✓
 - States that the induced electromotive force (emf) in a circuit equals the **rate of change of magnetic flux** through the circuit.
 - Correct.
3. **Lenz's Law and conservation of momentum** ✗
 - Incorrect. Lenz's Law is tied to the **law of conservation of energy**, ensuring that induced currents resist flux changes so energy is not created or destroyed.
 - It is **not** related to conservation of momentum.
4. **Transformer principle** ✓
 - Transformers operate on **mutual electromagnetic induction**, altering alternating voltage levels (step-up or step-down).
 - Correct.

Q. 45

Which of the following human hormones is **INCORRECTLY** paired with its primary physiological function?

- a) Insulin — Regulates blood glucose levels by promoting glucose uptake in tissues ✗
- b) Glucagon — Raises blood glucose concentration by promoting glycogenolysis in liver ✗
- c) Oxytocin — Stimulates uterine contractions during childbirth and milk ejection during lactation ✗
- d) Calcitonin — Elevates calcium ion concentrations in blood by stimulating bone breakdown ✓

✓ **Correct Answer:** Option D — Calcitonin: Elevates calcium ion concentrations in blood by stimulating bone breakdown

Explanation:

- **Insulin** ✓
 - Secreted by pancreas (β -cells).
 - Promotes glucose uptake in tissues, lowers blood sugar.
 - Correctly paired.
- **Glucagon** ✓
 - Secreted by pancreas (α -cells).
 - Stimulates glycogenolysis and gluconeogenesis in liver → raises blood glucose.
 - Correctly paired.
- **Oxytocin** ✓
 - Secreted by posterior pituitary.
 - Stimulates uterine contractions during childbirth and milk ejection during lactation.
 - Correctly paired.
- **Calcitonin** ✗
 - Secreted by thyroid gland (parafollicular cells).
 - **Function:** Lowers blood calcium levels by inhibiting bone resorption (osteoclast activity).
 - The statement given is **incorrect** because it says calcitonin elevates calcium by stimulating bone breakdown, which is the opposite of its actual role.
 - In reality, **parathyroid hormone (PTH)** elevates calcium by stimulating bone breakdown.

Explanation:

Q. 46

Match List-I (Major Rivers) with List-II (Their Key Tributaries):

River:	Tributaries:
A. Godavari	1. Bhima and Tungabhadra
B. Krishna	2. Indravati and Penganga
C. Cauvery	3. Seonath and Hasdeo
D. Mahanadi	4. Bhavani and Amravati

a) A-2, B-1, C-4, D-3



b) A-1, B-2, C-3, D-4



c) A-2, B-4, C-1, D-3



d) A-3, B-1, C-4, D-2



Explanation:

Correct Answer: Option A — A-2, B-1, C-4, D-3**Explanation:**

Let's match each major river with its key tributaries:

- **Godavari** → Tributaries include **Indravati, Penganga, Manjira, Sabari**.
◦ Correct match: **A → 2 (Indravati and Penganga)**
- **Krishna** → Tributaries include **Bhima, Tungabhadra, Ghataprabha, Malaprabha**.
◦ Correct match: **B → 1 (Bhima and Tungabhadra)**
- **Cauvery** → Tributaries include **Bhavani, Amravati, Hemavati, Kabini**.
◦ Correct match: **C → 4 (Bhavani and Amravati)**
- **Mahanadi** → Tributaries include **Seonath, Hasdeo, Jonk, Tel.**
◦ Correct match: **D → 3 (Seonath and Hasdeo)**

Q. 47

Consider the following statements regarding the administration of the Vijayanagara Empire:

1. The Nayankara system was an administrative structure where military chiefs (Nayakas) were granted territorial lands called Amaram.
2. Foreign traveler Nicolo de Conti visited Vijayanagara during the reign of King Deva Raya I.
3. Domingo Paes was a Portuguese traveler who recorded detailed observations during the rule of Krishnadevaraya.

Which of the statements given above are correct?

- a) 1 & 2 only 
- b) 2 & 3 only 
- c) 1 & 3 only 
- d) 1, 2 & 3 All 

 **Correct Answer:** Option D — 1, 2 & 3 All

Explanation:

1. **Nayankara system** 
 - Military chiefs called **Nayakas** were granted territorial lands known as **Amaram**.
 - In return, they maintained troops for the empire and paid tribute.
 - This was a crucial administrative and military arrangement under Vijayanagara rulers.
2. **Nicolo de Conti's visit** 
 - Italian traveler **Nicolo de Conti** visited Vijayanagara around **1420 CE**, during the reign of **Deva Raya I**.
 - He described the grandeur of the capital and the prosperity of the empire.
3. **Domingo Paes's account** 
 - Portuguese traveler **Domingo Paes** visited Vijayanagara in the early 16th century.
 - He recorded detailed observations during the reign of **Krishnadevaraya (1509–1529 CE)**.
 - His writings provide valuable insights into the empire's administration, economy, and military strength.

Explanation:

Q. 48

'Bhojali' folk dance is associated with which Indian State?

a) Chhattisgarh



b) Jharkhand



c) Odisha



d) Bihar



The folk dance 'Bhojali' is associated with the state of Chhattisgarh, where it is celebrated as part of a ritualistic friendship festival and cultural tradition.

Explanation:

- **Bhojali** is a traditional folk festival and dance form of **Chhattisgarh**, also observed in some parts of Madhya Pradesh.
- It is deeply tied to **ritualistic friendship bonds** among women, especially unmarried girls, who worship **Bhojali Dai (goddess)** by planting wheat or barley seedlings in baskets.
- The festival culminates in **songs (Bhojali geet)**, processions, and immersion of the seedlings in water bodies.
- The dance and songs symbolize **fertility, prosperity, and lifelong friendship bonds**.

Why Other Options Are Wrong:

- **Jharkhand**  → Known for tribal dances like **Chhau, Paika, Karma**, not Bhojali.
- **Odisha**  → Famous for **Gotipua, Odissi, Chhau (Mayurbhanj style)**, not Bhojali.
- **Bihar**  → Associated with **Jat-Jatin, Bidesia, Sohar-Khilouna**, not Bhojali.

Q. 49

Consider the following statements regarding the Comptroller and Auditor General (CAG) of India:

1. The CAG is appointed by the President by warrant under his hand and seal.
2. The administrative expenses of the office of the CAG are subject to the vote of Parliament.
3. The CAG holds office for a term of six years or until attaining sixty-five years of age, whichever is earlier.

Which of the statements given above are correct?

- a) 1 & 2 only 
- b) 2 & 3 only 
- c) 1 & 3 only 
- d) 1, 2 & 3 All 

 **Correct Answer:** Option C — 1 & 3 only

Explanation:

Let's evaluate each statement regarding the **Comptroller and Auditor General (CAG) of India**:

1. The CAG is appointed by the President by warrant under his hand and seal.
 -  Correct.
 - As per **Article 148**, the CAG is appointed by the President of India by warrant under his hand and seal.
2. The administrative expenses of the office of the CAG are subject to the vote of Parliament.
 -  Incorrect.
 - The administrative expenses of the CAG, including salaries, allowances, and pensions, are **charged upon the Consolidated Fund of India**.
 - Being "charged" means they are **not subject to parliamentary vote**, ensuring independence of the office.
3. The CAG holds office for a term of six years or until attaining sixty-five years of age, whichever is earlier.
 -  Correct.
 - Tenure: **6 years or up to age 65**, whichever is earlier.

Explanation:

Q. 50

Which of the following statements regarding Doppler Effect in light and sound is INCORRECT?

- a) Doppler effect in light waves is asymmetric and depends on absolute velocities through a medium. ✓
- b) Doppler effect in sound waves depends on whether the source or the observer is in physical motion. ✗
- c) Blue shift occurs when a light-emitting astronomical source moves towards an observer on Earth. ✗
- d) Red shift in astronomical spectra indicates that distant light sources are moving away from Earth. ✗

✓ **Correct Answer:** Option A — Doppler effect in light waves is asymmetric and depends on absolute velocities through a medium.

Explanation

- The **Doppler Effect** describes the change in frequency/wavelength of waves due to relative motion between source and observer.

For Sound Waves

- Medium is required.
- The effect depends on whether the **source** or the **observer** is moving.
- Example: Ambulance siren sounds higher-pitched when approaching, lower-pitched when receding.

Explanation:

- ✓ Statement 2 is **correct**.

For Light Waves

- Light is an **electromagnetic wave**; it does not require a medium.
- Doppler effect in light depends only on **relative velocity** between source and observer, not on absolute velocities.
- Blue shift** → Source moves **towards** observer → wavelength decreases.
- Red shift** → Source moves **away** from observer → wavelength increases.
- ✓ Statements 3 and 4 are **correct**.
- ✗ Statement 1 is **incorrect**, because Doppler effect in light is **symmetric** and depends only on relative motion, not absolute velocities through a medium.

Q. 51	Rearrange the following sentences to form a meaningful paragraph. A. Therefore, many companies have started implementing strict cybersecurity measures to protect their confidential information from potential threats. B. The increasing dependence on digital platforms has created new challenges related to data security and online privacy. C. These security concerns have become more serious due to the frequent occurrence of cyberattacks targeting individuals and organisations. D. However, specialists believe that regular updates, employee awareness, and responsible online behaviour are equally important for maintaining a secure digital environment. a) C-B-A-D  b) B-A-D-C  c) A-B-C-D  d) B-C-A-D 
Explanation:	Ans: (d) B-C-A-D Solution: • B is the opening sentence because it introduces the main topic "increasing dependence on digital platforms and security challenges". • C follows B because "These security concerns" refers to the data security issues mentioned in the previous sentence. • A comes next because "Therefore" indicates the action taken as a result of these concerns. • D is the concluding sentence because "However" introduces an additional important point related to cybersecurity. Hence, the correct order is B-C-A-D.

Q. 52	Choose the correct meaning of the idiom: "To toe the line" a) To oppose someone's decision openly  b) To follow rules, orders, or accepted standards strictly  c) To create unnecessary problems for others  d) To take a difficult decision without support 
Explanation:	Ans: (b) To follow rules, orders, or accepted standards strictly Solution: The idiom "To toe the line" means to obey rules, instructions, or authority without opposition. To toe the line = To follow rules / To comply with instructions हिंदी: To toe the line का अर्थ है नियमों या आदेशों का पालन करना और निर्धारित तरीके से काम करना। • To toe the line → नियमों का पालन करना / आदेश मानना • Oppose → विरोध करना • Create problems → समस्याएँ उत्पन्न करना

Q. 53	Select the incorrectly spelt word. a) Scintilators ✓ b) Miscellaneous ✗ c) Conscientious ✗ d) Accommodate ✗
Explanation:	Ans: (a) Scintilators Solution: The incorrectly spelt word is "Scintilators". The correct spelling is "Scintillators" (with double "l"). Scintillators are materials that produce flashes of light when exposed to radiation or high-energy particles. हिंदी: गलत spelling वाला शब्द Scintilators है। इसका सही spelling Scintillators होता है जिसमें "ll" (double l) आता है। • Scintillators ✓ (सही) • Scintilators ✗ (गलत)

Q. 54	Identify the segment in the sentence which contains a grammatical error. The company's data, / which has been collected and analyzed / for over a decade, / support the conclusion that the new strategy is ineffective. a) The company's data ✗ b) which has been collected and analyzed ✗ c) for over a decade ✗ d) support the conclusion that the new strategy is ineffective ✓
Explanation:	Ans: (d) support the conclusion that the new strategy is ineffective Solution: The error is in part (d). Here, the word "data" is used as a singular collective noun referring to a collection of information. Therefore, the verb should be used in the singular form. support → supports Correct Sentence: The company's data, which has been collected and analyzed for over a decade, supports the conclusion that the new strategy is ineffective. हिंदी: यहाँ data को एक collective noun (एकत्रित जानकारी) के रूप में प्रयोग किया गया है, इसलिए इसके साथ singular verb का प्रयोग होगा। support (plural verb) की जगह supports (singular verb) आएगा।

Q. 55

Choose the correct Passive Voice of the given Active Voice:

The committee has approved the proposal and the authorities are implementing the new policy in several departments.

- a) The proposal was approved by the committee and the new policy has been implemented by the authorities in several departments. 
- b) The proposal has been approved by the committee and the new policy is being implemented by the authorities in several departments. 
- c) The proposal had been approved by the committee and the new policy was implemented by the authorities in several departments. 
- d) The proposal has been approved by the committee and the new policy is being implemented by the authority in several departments. 

Ans: (b) The proposal has been approved by the committee and the new policy is being implemented by the authorities in several departments.

Solution:

The given sentence contains **two different tenses**, so both parts will change separately while converting into Passive Voice.

First part:

Active: The committee **has approved** the proposal.

- Tense: Present Perfect
- Structure: has/have + V³
- Passive: has/have been + V³

has approved → has been approved

Second part:

Active: The authorities **are implementing** the new policy.

- Tense: Present Continuous
- Structure: is/am/are + V¹-ing
- Passive: is/am/are + being + V³

are implementing → is being implemented

Explanation:

Q. 56

Q. Select the incorrectly spelt word.

- a) Remunative ✓
- b) Rehearsal ✗
- c) Reimbursement ✗
- d) Significance ✗

Ans: (a) Remunative

Solution:

The incorrectly spelt word is "Remunative".

The correct spelling is "Remunerative".

Meaning of all options:

Explanation:

- **Remunative** ✗ → Incorrect spelling. Correct spelling is **Remunerative**.
Remunerative → Providing good payment, profit, or financial reward.
(लाभदायक / अच्छा पारिश्रमिक देने वाला)
- **Rehearsal** ✓ → A practice session before a performance or event.
(अभ्यास / पूर्वाभ्यास)
- **Reimbursement** ✓ → Repayment of money spent by someone for an official purpose.
(खर्च की गई राशि की वापसी / प्रतिपूर्ति)
- **Significance** ✓ → Importance or meaning of something.
(महत्व / अर्थ)

Q. 57

Find the part of the sentence that contains an error:

Perhaps the biggest threat today (a) /
is the impact of climate change on food (b) /
systems and livelihoods of poor farmers. (c) /
No error (d)

a) a b) b c) c d) d 

Ans: (d) No error

Solution:

The sentence is grammatically correct. There is **no error** in any part.

Explanation:

- "the biggest threat" → correct use of superlative degree.
- "impact of climate change on food systems" → correct preposition usage.
- "livelihoods of poor farmers" → grammatically correct phrase.

Explanation:

हिंदी:

इस वाक्य में कोई grammatical error नहीं है।

- **Impact on** का प्रयोग सही है क्योंकि किसी चीज़ के प्रभाव के लिए **impact on** का प्रयोग किया जाता है।
- **Food systems and livelihoods** दोनों nouns सही रूप में प्रयोग हुए हैं।

Correct Sentence:

Perhaps the biggest threat today is the impact of climate change on food systems and livelihoods of poor farmers.

Q. 58	Choose the correct meaning of the phrasal verb "Cross out" a) To add something extra  b) To remove something by drawing a line  c) To copy something carefully  d) To search for something 
Explanation:	Ans: (b) To remove or cancel something by drawing a line through it Solution: The phrasal verb "Cross out" means to remove, delete, or cancel a word, sentence, or item by drawing a line over it. Cross out = Delete / Cancel / Remove (काट देना या हटाना)

Q. 59

Choose the correct Indirect Speech of the given Direct Speech:

The teacher said to the students, "Why have you not completed your assignments despite being given sufficient time?"

- a) The teacher asked the students why they had not completed their assignments despite being given sufficient time. 
- b) The teacher asked the students why they have not completed their assignments despite being given sufficient time. 
- c) The teacher said to the students why had they not completed their assignments despite being given sufficient time. 
- d) The teacher asked the students why they did not complete their assignments despite being given sufficient time. 

Ans: (a) The teacher asked the students why they had not completed their assignments despite being given sufficient time.

Solution:

The sentence is an **interrogative sentence** beginning with "Why", so in indirect speech **said to** changes into **asked**.

Rules applied:

Explanation:

- Question form changes into statement form.
- Have completed → had completed (Present Perfect changes into Past Perfect).
- You → they according to the reporting object.

हिंदी:

यह WH-question है, इसलिए indirect speech में **said to** की जगह **asked** आएगा।

Present Perfect (**have completed**) को Past Perfect (**had completed**) में बदला जाएगा।

Q. 60

Choose the correct option to fill in the blank:

The scientist was deeply absorbed _____ his research work and did not notice the disturbance around him.

- a) for ✗
b) on ✗
c) with ✗
d) in ✓

Ans: (d) in

Solution:

The correct phrase is "**absorbed in**", which means **completely involved or interested in something**.

Explanation:

Fixed Preposition:

Absorbed in → Completely involved in something (किसी कार्य में पूरी तरह डूबा होना)

हिंदी:

वाक्य में वैज्ञानिक अपने research work में पूरी तरह व्यस्त था, इसलिए **absorbed in** का प्रयोग सही होगा।

- **Absorbed in** → में लीन होना / पूरी तरह व्यस्त होना
- **Absorbed with/on/for** → गलत प्रयोग

Q. 61	Rearrange the following sentences to form a meaningful paragraph. A. As a result, several governments have introduced new policies to encourage the use of renewable sources of energy and reduce environmental damage. B. The continuous rise in pollution levels has become a serious concern for countries around the world in recent years. C. These initiatives have not only promoted sustainable development but have also increased awareness among people regarding environmental protection. D. However, environmental experts believe that long-term solutions require collective efforts from governments, industries, and individuals. a) B-A-C-D ✓ b) A-B-C-D ✗ c) B-C-A-D ✗ d) C-B-D-A ✗
Explanation:	Ans: (a) B-A-C-D Solution: • B is the opening sentence because it introduces the main topic "rise in pollution levels". • A follows B because "As a result" shows the actions taken due to increasing pollution. • C comes after A because "These initiatives" refers to the policies mentioned in A. • D is the concluding sentence as "However" introduces the final opinion of experts. Hence, the correct order is B-A-C-D.

Q. 62	Select the most appropriate option to substitute the underlined segment in the given sentence. The young boy takes after his father in terms of his intelligence and problem-solving abilities. a) resembles his father ✓ b) takes care of his father ✗ c) depends on his father ✗ d) follows his father ✗
Explanation:	Ans: (a) resembles his father Solution: The phrasal verb "take after" means to resemble a family member, especially in appearance, behaviour, or qualities. यहाँ takes after his father का अर्थ है कि लड़का अपने पिता जैसा दिखता है या उनके गुणों को अपनाता है। • Take after = Resemble (समान होना / जैसा होना) • Take care of = Look after (देखभाल करना) • Depend on = Rely on (निर्भर होना) अतः सही उत्तर: (a) resembles his father.

Q. 63

Choose the correct one-word substitution for the given phrase:

The commercial rearing of silkworms for the production of silk is known as:

- a) Apiculture 
- b) Silviculture 
- c) Sericulture 
- d) Pisciculture 

Ans: (c) Sericulture

Solution:

The correct answer is **Sericulture**.

Sericulture means the commercial breeding and rearing of silkworms for the production of silk.

हिंदी:

Explanation:

Sericulture (रेशमकीट पालन) का अर्थ है रेशम प्राप्त करने के लिए रेशम के कीड़ों का व्यावसायिक पालन करना।

- Sericulture → Rearing of silkworms for silk production (रेशम उत्पादन हेतु रेशमकीट पालन)
- Pisciculture → Breeding of fish (मत्स्य पालन)
- Silviculture → Forest cultivation/management (वनों का प्रबंधन एवं संवर्धन)
- Apiculture → Bee keeping (मधुमक्खी पालन)

Q. 64	Choose the correct Direct Speech of the given Indirect Speech: The scientist explained that the experiment had failed because the required conditions were not available at that time. a) The scientist said, "The experiment failed because the required conditions are not available at that time."  b) The scientist said, "The experiment had failed because the required conditions were not available at that time."  c) The scientist said, "The experiment has failed because the required conditions were not available at that time."  d) The scientist said, "The experiment fails because the required conditions were not available at that time." 
Explanation:	Ans: (b) The scientist said, "The experiment had failed because the required conditions were not available at that time." Solution: While changing indirect speech into direct speech, the original tense of the speaker's words is restored. Here, the reporting sentence contains had failed and were not available, which indicate a past situation. Therefore, the correct direct form will retain the same tense. हिंदी: Indirect से Direct Speech में बदलते समय वक्ता के वास्तविक शब्द वापस लाए जाते हैं। यहाँ घटना भूतकाल से संबंधित है, इसलिए had failed और were not available का प्रयोग सही रहेगा।

Q. 65

Choose the correct option to fill in the blank:

The committee recommended that the new policy _____ implemented immediately to avoid further complications.

- a) is ✗
b) was ✗
c) be ✓
d) has been ✗

Ans: (c) be

Solution:

The correct answer is "be".

After verbs like **recommend, suggest, demand, insist, propose**, we use the **subjunctive mood**, where the base form of the verb is used.

Rule:

Subject + recommended/suggested + that + subject + V¹ (base form)

Explanation:

Here:

recommended that the new policy be implemented ✓

हिंदी:

जब **recommend, suggest, demand, insist, propose** जैसे verbs के बाद **that clause** आता है, तो verb की **first form (base form)** का प्रयोग किया जाता है।

इसलिए:

- is implemented ✗
- was implemented ✗
- be implemented ✓

Q. 66

Select the most appropriate option to substitute the underlined segment in the given sentence.

He did not like me to smoking in the presence of our teacher yesterday.

- a) that I smoke 
- b) my smoking 
- c) me smoking 
- d) No correction required 

Ans: (b) my smoking

Solution:

The correct answer is "my smoking".

After verbs like **like**, **dislike**, **hate**, **love**, **enjoy**, **avoid**, when we talk about an activity, we generally use a **gerund** (V¹ + ing).

Here, **smoking** is an activity, so it should be preceded by a possessive form (**my**) instead of the object form (**me**).

Explanation:

Rule:

Possessive adjective + Gerund (V¹ + ing)

Examples:

- I dislike **his talking** loudly. 
- She enjoys **my singing**. 

हिंदी:

जब किसी कार्य (activity) को बताने के लिए **verb + ing** का प्रयोग होता है, तो उसके पहले possessive form (my, his, her, their) का प्रयोग किया जाता है।

Q. 67

Choose the correct one-word substitution for the given phrase:

The state of perfect peace and freedom from all suffering, according to Buddhist philosophy, is called:

a) Meditation

b) Salvation

c) Nirvana

d) Enlightenment

Ans: (c) Nirvana

Solution:

The correct answer is **Nirvana**.

Nirvana refers to a state of complete peace, liberation, and freedom from the cycle of birth and death in **Buddhist philosophy**.

Explanation:

हिंदी:

Nirvana (निर्वाण) का अर्थ है पूर्ण शांति और दुखों से मुक्ति की अवस्था। बौद्ध धर्म के अनुसार, निर्वाण वह अवस्था है जहाँ व्यक्ति जन्म और मृत्यु के चक्र से मुक्त हो जाता है।

- Nirvana → A state of complete peace and liberation (पूर्ण मुक्ति की अवस्था)
- Salvation → मुक्ति/उद्धार
- Meditation → ध्यान
- Enlightenment → ज्ञान प्राप्ति

Q. 68	Choose the correct synonym of the word "Felicity": a) Serenity  b) Happiness  c) Prosperity  d) Satisfaction 
Explanation:	Ans: (b) Happiness Solution: The correct synonym of "Felicity" is "Happiness". Felicity means a state of intense happiness, great joy, or pleasure. Among the given options, happiness is the closest meaning. हिंदी: Felicity का अर्थ है अत्यधिक खुशी, आनंद या प्रसन्नता। • Felicity → Happiness / Great joy (अत्यधिक खुशी) • Serenity → Peace and calmness (शांति / स्थिरता) • Prosperity → Wealth and success (समृद्धि) • Satisfaction → Contentment (संतोष)

Q. 69	Choose the correct antonym of the word "Impetuous": a) Rash  b) Hasty  c) Cautious  d) Reckless 
Explanation:	Ans: (c) Cautious Solution: The correct antonym of "Impetuous" is "Cautious". Impetuous means acting suddenly without thinking about the consequences; showing impulsive behaviour. Therefore, its opposite is Cautious, which means thinking carefully before taking any action. हिंदी: Impetuous का अर्थ है बिना सोचे-समझे जल्दबाजी में कार्य करने वाला या आवेगपूर्ण। इसका विपरीत अर्थ Cautious (सावधान और सोच-समझकर कार्य करने वाला) होता है। Meaning of all options: • Impetuous → Impulsive / Acting without thinking (आवेगपूर्ण / बिना सोचे कार्य करने वाला) • Rash → Acting too quickly and carelessly (लापरवाह तरीके से जल्दबाजी करने वाला) • Hasty → Done too quickly without enough thought (जल्दबाजी में किया गया) • Cautious → Careful and avoiding risks (सावधान / सतर्क) • Reckless → Showing no concern for danger or consequences (लापरवाह / जोखिम की परवाह न करने वाला)

Q. 70	Choose the incorrect option regarding the use of the word "Drop": a) The temperature began to drop as the evening progressed. ✗ b) He accidentally dropped his mobile phone on the floor. ✗ c) The teacher asked the students to drop their doubts before the examination. ✓ d) The old man's health started to drop after the surgery. ✗
Explanation:	Ans: (c) The teacher asked the students to drop their doubts before the examination. Solution: The word "Drop" has different meanings, but it cannot be used in every situation. • Drop (to fall/decrease) → सही प्रयोग <i>The temperature began to drop.</i> ✓ • Drop (to let something fall accidentally) → सही प्रयोग <i>He dropped his mobile phone.</i> ✓ • Drop (to decrease in condition/level) → सही प्रयोग <i>Health started to drop.</i> ✓ Drop doubts ✗ is incorrect because doubts are not physically dropped. The correct expression is "clear doubts" or "remove doubts".

Q. 71

Read the following passage carefully and answer the questions given below:

In recent years, the concept of sustainable development has gained significant importance across the world. Rapid industrialisation and excessive use of natural resources have created several environmental challenges, including climate change, pollution, and loss of biodiversity. To address these issues, governments, organisations, and individuals are increasingly focusing on adopting eco-friendly practices that can ensure a balance between economic growth and environmental protection.

Renewable energy sources such as solar, wind, and hydropower have emerged as effective alternatives to traditional sources of energy. These resources not only reduce dependence on fossil fuels but also help in decreasing harmful emissions. However, the transition towards sustainable energy requires proper planning, advanced technology, and cooperation among different sectors of society.

Although many countries have introduced policies to promote sustainability, their successful implementation remains a challenge. Public awareness and responsible behaviour play a crucial role in achieving long-term environmental goals. It is essential that people understand the importance of conserving resources and make conscious choices in their daily lives.

The future of the planet depends on collective efforts to create a sustainable environment. If governments and citizens work together with determination, significant progress can be made in protecting natural resources for future generations.

What is the tone of the passage?

- a) Humorous and entertaining 
- b) Critical and negative 
- c) Informative and concerned 
- d) Casual and indifferent 

Explanation:

Ans: (c) Informative and concerned

Solution:

The writer provides information about environmental issues and also expresses concern about protecting the planet. Therefore, the tone is **informative and concerned**.

हिंदी:

लेखक पर्यावरण संबंधी समस्याओं की जानकारी देने के साथ-साथ उनके समाधान की चिंता भी व्यक्त करता है, इसलिए tone **Informative and concerned** है।

Q. 72

Read the following passage carefully and answer the questions given below:

In recent years, the concept of sustainable development has gained significant importance across the world. Rapid industrialisation and excessive use of natural resources have created several environmental challenges, including climate change, pollution, and loss of biodiversity. To address these issues, governments, organisations, and individuals are increasingly focusing on adopting eco-friendly practices that can ensure a balance between economic growth and environmental protection.

Renewable energy sources such as solar, wind, and hydropower have emerged as effective alternatives to traditional sources of energy. These resources not only reduce dependence on fossil fuels but also help in decreasing harmful emissions. However, the transition towards sustainable energy requires proper planning, advanced technology, and cooperation among different sectors of society.

Although many countries have introduced policies to promote sustainability, their successful implementation remains a challenge. Public awareness and responsible behaviour play a crucial role in achieving long-term environmental goals. It is essential that people understand the importance of conserving resources and make conscious choices in their daily lives.

The future of the planet depends on collective efforts to create a sustainable environment. If governments and citizens work together with determination, significant progress can be made in protecting natural resources for future generations.

What is the main theme of the passage?

- a) The impact of industrialisation on economic growth 
- b) The importance of sustainable development and environmental protection 
- c) The disadvantages of renewable energy sources 
- d) The role of technology in increasing pollution 

Explanation:

Ans: (b) The importance of sustainable development and environmental protection

Solution:

The passage mainly discusses the need for sustainable development, the challenges caused by environmental damage, and the importance of protecting natural resources.

हिंदी:

इस passage का मुख्य विषय **सतत विकास और पर्यावरण संरक्षण का महत्व** है।

Q. 73

Read the following passage carefully and answer the questions given below:

In recent years, the concept of sustainable development has gained significant importance across the world. Rapid industrialisation and excessive use of natural resources have created several environmental challenges, including climate change, pollution, and loss of biodiversity. To address these issues, governments, organisations, and individuals are increasingly focusing on adopting eco-friendly practices that can ensure a balance between economic growth and environmental protection.

Renewable energy sources such as solar, wind, and hydropower have emerged as effective alternatives to traditional sources of energy. These resources not only reduce dependence on fossil fuels but also help in decreasing harmful emissions. However, the transition towards sustainable energy requires proper planning, advanced technology, and cooperation among different sectors of society.

Although many countries have introduced policies to promote sustainability, their successful implementation remains a challenge. Public awareness and responsible behaviour play a crucial role in achieving long-term environmental goals. It is essential that people understand the importance of conserving resources and make conscious choices in their daily lives.

The future of the planet depends on collective efforts to create a sustainable environment. If governments and citizens work together with determination, significant progress can be made in protecting natural resources for future generations.

According to the passage, why are renewable energy sources considered effective alternatives?

- a) They increase the use of fossil fuels 
- b) They prevent all environmental problems immediately 
- c) They completely eliminate the need for technology 
- d) They reduce harmful emissions and dependence on traditional energy sources 

Ans: (d) They reduce harmful emissions and dependence on traditional energy sources

Solution:

The passage states that renewable energy sources help reduce dependence on fossil fuels and decrease harmful emissions.

Explanation:

हिंदी:

Renewable energy sources जीवाश्म ईंधन पर निर्भरता कम करते हैं और हानिकारक emissions को घटाते हैं।

Q. 74

Read the following passage carefully and answer the questions given below:

In recent years, the concept of sustainable development has gained significant importance across the world. Rapid industrialisation and excessive use of natural resources have created several environmental challenges, including climate change, pollution, and loss of biodiversity. To address these issues, governments, organisations, and individuals are increasingly focusing on adopting eco-friendly practices that can ensure a balance between economic growth and environmental protection.

Renewable energy sources such as solar, wind, and hydropower have emerged as effective alternatives to traditional sources of energy. These resources not only reduce dependence on fossil fuels but also help in decreasing harmful emissions. However, the transition towards sustainable energy requires proper planning, advanced technology, and cooperation among different sectors of society.

Although many countries have introduced policies to promote sustainability, their successful implementation remains a challenge. Public awareness and responsible behaviour play a crucial role in achieving long-term environmental goals. It is essential that people understand the importance of conserving resources and make conscious choices in their daily lives.

The future of the planet depends on collective efforts to create a sustainable environment. If governments and citizens work together with determination, significant progress can be made in protecting natural resources for future generations.

Identify the clause in the following sentence from the passage:

The underlined part "that people understand the importance of conserving resources" is:

- a) Adjective Clause 
- b) Adverb Clause 
- c) Noun Clause 
- d) Principal Clause 

Ans: (c) Noun Clause

Solution:

The correct answer is **Noun Clause**.

The clause "**that people understand the importance of conserving resources**" acts as a **noun** because it represents a fact/idea that is considered essential.

Explanation:

Sentence structure:

It is essential + that people understand the importance of conserving resources.

Here, the clause answers the question "**What is essential?**"

→ **That people understand the importance of conserving resources.**

Therefore, it functions as a **Noun Clause**.

Q. 75

Read the following passage carefully and answer the questions given below:

In recent years, the concept of sustainable development has gained significant importance across the world. Rapid industrialisation and excessive use of natural resources have created several environmental challenges, including climate change, pollution, and loss of biodiversity. To address these issues, governments, organisations, and individuals are increasingly focusing on adopting eco-friendly practices that can ensure a balance between economic growth and environmental protection.

Renewable energy sources such as solar, wind, and hydropower have emerged as effective alternatives to traditional sources of energy. These resources not only reduce dependence on fossil fuels but also help in decreasing harmful emissions. However, the transition towards sustainable energy requires proper planning, advanced technology, and cooperation among different sectors of society.

Although many countries have introduced policies to promote sustainability, their successful implementation remains a challenge. Public awareness and responsible behaviour play a crucial role in achieving long-term environmental goals. It is essential that people understand the importance of conserving resources and make conscious choices in their daily lives.

The future of the planet depends on collective efforts to create a sustainable environment. If governments and citizens work together with determination, significant progress can be made in protecting natural resources for future generations.

Which of the following statements is TRUE according to the passage?

- a) Sustainable development focuses only on economic growth. 
- b) Public awareness has no role in environmental protection. 
- c) Cooperation among different sectors is necessary for sustainable energy transition. 
- d) Renewable energy sources increase environmental damage. 

Explanation:

Ans: (c) Cooperation among different sectors is necessary for sustainable energy transition.

Solution:

The passage clearly mentions that the transition towards sustainable energy requires cooperation among different sectors of society.

हिंदी:

Passage के अनुसार sustainable energy के लिए समाज के विभिन्न क्षेत्रों के बीच सहयोग आवश्यक है।

Q. 76

If $x^2 + 4x + 1 = 0$, then what is the value of $x^3 + \frac{1}{x^3}$?

a) 52 

b) -54 

c) -52 

d) 54 

Explanation:

Given:

$$x^2 + 4x + 1 = 0$$

Divide the whole equation by x:

$$x + 4 + \frac{1}{x} = 0$$

$$x + \frac{1}{x} = -4$$

Formula used:

$$a^3 + b^3 = (a + b)^3 - 3ab(a + b)$$

Here,

$$a = x, b = \frac{1}{x}$$

$$ab = x \times \frac{1}{x} = 1$$

Therefore,

$$x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x\right)\left(\frac{1}{x}\right)\left(x + \frac{1}{x}\right)$$

$$= (-4)^3 - 3(1)(-4)$$

$$= -64 + 12 = -52$$

$$\text{Answer} = -52$$

Q. 77

ABCD is a quadrilateral with $\angle A = 3(x + 2)^\circ$, $\angle B = 2(2x - 7)^\circ$, $\angle C = 69^\circ$ and $\angle D = \angle C - 8^\circ$. What is the value of $\angle B$?

a) 183° b) 128° c) 122° d) 153° **Explanation:**

$$\angle C = 69^\circ$$

$$\angle D = 69^\circ - 8^\circ = 61^\circ$$

Sum of angles of a quadrilateral = 360°

$$3(x + 2) + 2(2x - 7) + 69 + 61 = 360$$

$$3x + 6 + 4x - 14 + 130 = 360$$

$$7x + 122 = 360$$

$$7x = 238$$

$$x = 34$$

Now,

$$\angle B = 2(2x - 7) = 2(68 - 7) = 2 \times 61 = 122^\circ$$

Answer: C) 122°

Q. 78

Divide 25 into two parts such that 6 times of the larger part added to 4 times of the smaller part is equal to 130. Find the larger part.

a) 13 b) 20 c) 15 d) 14 **Explanation:**Let larger part = x Smaller part = $25 - x$

Given:

$$6x + 4(25 - x) = 130$$

$$\Rightarrow 6x + 100 - 4x = 130$$

$$\Rightarrow 2x + 100 = 130$$

$$\Rightarrow 2x = 30$$

$$\Rightarrow x = 15$$

Answer: (c) 15

Q. 79

Varun bought some erasers at the rate of ₹150/dozen. He sold them for ₹18 each. His profit percentage is ____%.

- a) 46 
- b) 44 
- c) 42 
- d) 48 

Explanation:

CP of 12 erasers = ₹150

CP of 1 eraser = $150/12 = ₹12.50$

SP of 1 eraser = ₹18

Profit = SP - CP = $18 - 12.50 = ₹5.50$

Profit % = $(\text{Profit}/\text{CP}) \times 100$

= $(5.50/12.50) \times 100$

= 44%

Answer: B) 44%

Q. 80

A and B, when each is working alone, can complete a piece of work in, respectively, 20 days and 25 days. Each works on alternate days, with A and B starting to work on Day 1 and Day 2, respectively. How long will it take them to complete the work?

- a) 22 $\frac{1}{5}$ days 
- b) 23 days 
- c) 11 days 
- d) 22 days 

Explanation:

Take total work = LCM(20, 25) = 100 units

$$A's \text{ 1-day work} = \frac{100}{20} = 5 \text{ units}$$

$$B's \text{ 1-day work} = \frac{100}{25} = 4 \text{ units}$$

In 2 days, work done by A + B:

$$= 5 + 4 = 9 \text{ units}$$

In 22 days (11 pairs):

$$\text{Work done} = 11 \times 9 = 99 \text{ units}$$

$$\text{Remaining work} = 100 - 99 = 1 \text{ unit}$$

Day 23 is A's turn.

A does 5 units in 1 day.

$$\text{Time to do 1 unit} = \frac{1}{5} \text{ day}$$

$$\text{Therefore, total time:} = 22 + \frac{1}{5} = 22 \frac{1}{5} \text{ days}$$

Answer: A) 22 ($\frac{1}{5}$) days

Q. 81

A number when increased by 100% gives 2770. The number is:

- a) 4155 
- b) 1385 
- c) 692.5 
- d) 2770 

Explanation:

Original number = 100%

Increased by 100%:

$$100\% + 100\% = 200\%$$

So,

$$200\% = 2770$$

$$100\% = 2770/2 = 1385$$

Answer: B) 1385

Q. 82

If $\sec A + \tan A = 5$, then $\sec A - \tan A = ?$

- a) $5/2$ 
- b) $1/5$ 
- c) 5 
- d) $2/5$ 

Explanation:

$$(\sec A + \tan A)(\sec A - \tan A) = 1$$

Given:

$$\sec A + \tan A = 5$$

Therefore,

$$5(\sec A - \tan A) = 1$$

$$\sec A - \tan A = 1/5$$

Answer: B) $1/5$

Q. 83

Find the simple interest (in ₹) on ₹1,200 at an interest rate of 4% per month for 4 months.

a) ₹36 b) ₹32 c) ₹192 d) ₹16 **Explanation:**

$$SI = \frac{P \times R \times T}{100}$$

Here,

P = ₹1200, R = 4% per month, T = 4 months

$$SI = \frac{1200 \times 4 \times 4}{100}$$

$$= 12 \times 16 = ₹192$$

Answer: C) ₹192

Q. 84

A trader claims to sell flour at a profit of 20%, but also dishonestly uses a weight which is 10% less than what is mentioned on it. Find the total percentage of profit earned by the trader. (Round the answer to the nearest whole number.)

a) 33% b) 32% c) 35% d) 36% **Explanation:**

Let CP of 1 kg flour = ₹100

At 20% profit:

SP = ₹120

But he gives 10% less weight:

Actual quantity given = 90% of 1 kg

Cost of actual quantity = ₹90

Formula used:

$$\text{Profit \%} = (\text{Profit}/\text{CP}) \times 100$$
$$\text{Profit} = 120 - 90 = ₹30$$
$$\text{Profit \%} = (30/90) \times 100 = 33.33\%$$
$$\approx 33\%$$

Answer: 33%

Q. 85

The average age of Mohan and his four friends is 36 years. Mohan is 5 years older than the average age of his four friends. Determine Mohan's age in years.

- a) 42 
- b) 40 
- c) 44 
- d) 38 

Explanation:

let the average age of 4 friends = x years.

Mohan's age = $x + 5$

Total age of all 5 persons = $36 \times 5 = 180$

So,

$$4x + (x + 5) = 180$$

$$5x + 5 = 180$$

$$5x = 175$$

$$x = 35$$

Therefore,

$$\text{Mohan's age} = 35 + 5 = \mathbf{40 \text{ years}}$$

Answer: B) 40 years

Q. 86

The radius of the base of a right circular cone is 7 cm and its vertical height is 24 cm. Find the Volume of the cone.

a) 1133 cm³ b) 1232 cm³ c) 1331 cm³ d) 1330 cm³ **Explanation:**

Radius (r) = 7 cm

Height (h) = 24 cm

Formula: Volume of cone = $\frac{1}{3} \times \pi r^2 h$

$$= \frac{1}{3} \times \frac{22}{7} \times 7 \times 7 \times 24$$

$$= 22 \times 7 \times 8 = 1232 \text{ cm}^3$$

Answer = 1232 cm³ (Option B)

Q. 87

A man driving a car at a speed of 54 km/hr crosses a bridge in 1.7 minutes. Find the length of the bridge.

a) 1.53 km b) 2.58 km c) 2.13 km d) 0.92 km **Explanation:**Time = 1.7 min = $1.7/60$ hrDistance = Speed $\times$ Time

$$\text{Length of bridge} = 54 \times \frac{1.7}{60} = 0.9 \times 1.7$$

$$= 1.53 \text{ km}$$

Answer: A) 1.53 km

Q. 88

In what ratio should rice costing ₹40 per kg be mixed with rice costing ₹60 per kg so that the mixture costs ₹48 per kg?

- a) 3 : 2 
- b) 1 : 2 
- c) 2 : 1 
- d) 3 : 1 

Using Alligation:

$$\begin{array}{r} \text{₹40} \qquad \text{₹60} \\ \backslash \qquad / \\ \text{₹48} \\ / \qquad \backslash \\ 60 - 48 \quad 48 - 40 \\ 12 \qquad 8 \end{array}$$

Explanation:

Therefore,

$$\begin{aligned} \text{₹40 rice : ₹60 rice} \\ = 12 : 8 = 3 : 2 \end{aligned}$$

Answer: A) 3 : 2

Q. 89

P and Q are two stations. A train travels from P to Q at a speed of 144 km/h and returns to P at a slower speed. If the average speed for the entire journey is 126 km/h, find the speed at which the train returns.

a) 124 km/h



b) 128 km/h



c) 121 km/h



d) 112 km/h

**Explanation:**

For equal distances,

Formula:

$$\text{Average Speed} = \frac{2xy}{x+y}$$

Let the return speed = x km/h.

$$126 = \frac{2 \times 144 \times x}{144 + x}$$

$$126(144 + x) = 288x$$

$$18144 + 126x = 288x$$

$$162x = 18144$$

$$x = 112 \text{ km/h}$$

Therefore, the return speed = 112 km/h.

Answer: D) 112 km/h

Q. 90

The outer and inner radius of a hollow hemispherical shell are 21 cm and 14 cm respectively. Find the approximate volume of the metal used.

a) 13445 cm^3



b) 15675 cm^3



c) 12425 cm^3



d) 13655 cm^3

**Explanation:**

$$\text{Volume of hemisphere} = \left(\frac{2}{3}\right)\pi r^3$$

$$\text{Required volume} = \left(\frac{2}{3}\right)\pi(R^3 - r^3)$$

$$= \left(\frac{2}{3}\right)\pi(21^3 - 14^3)$$

$$= \left(\frac{2}{3}\right)\pi(9261 - 2744)$$

$$= \left(\frac{2}{3}\right)\pi(6517)$$

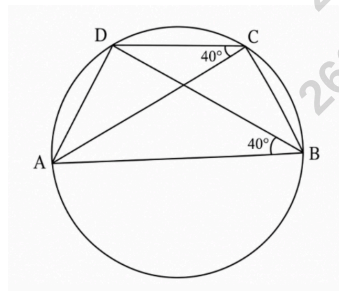
$$= \frac{2 \times 6517 \times 22}{3 \times 7}$$

$$= \frac{13034 \times 22}{21}$$

$$= \frac{286748}{21} = 13655 \text{ cm}^3$$

Q. 91

In a cyclic quadrilateral ABCD, AB is the diameter of the circle. If angle ACD is 40° , what is the measure of angle BAD?

a) 40° ☒b) 50° ☒c) 80° ☒d) 90° ☒**Explanation:** $\angle ABD = \angle ACD$ (on same chord AD)

Now,

 $\angle ADB = 90^\circ$ ($\because$ AB is diameter)So, in $\triangle ADB$

$$\angle BAD + 90^\circ + 40^\circ = 180^\circ$$

$$\angle BAD + 130^\circ = 180^\circ$$

$$\angle BAD = 180^\circ - 130^\circ$$

$$\angle BAD = 50^\circ$$

Q. 92

Which of the following is the greatest number that, when dividing 183, 127 and 211, leaves the same remainder each time?

- a) 32 
- b) 25 
- c) 30 
- d) 28 

Explanation:

When the same remainder is left, take the **HCF of the differences**.

$$183 - 127 = 56$$

$$211 - 183 = 28$$

$$211 - 127 = 84$$

$$\text{HCF}(56, 28, 84) = 28$$

Therefore, the greatest required number = 28

Answer: D) 28

Q. 93

If the sum of all interior angles of a regular polygon is 28 right angles, then its number of sides is:

a) 16 

b) 15 

c) 13 

d) 14 

Explanation:

1 right angle = 90°

Sum of interior angles = $28 \times 90^\circ = 2520^\circ$

Sum of interior angles = $(n - 2) \times 180^\circ$

So,

$(n - 2) \times 180^\circ = 2520^\circ$

$n - 2 = 14$

$n = 16$

Answer: A) 16

Q. 94

Evaluate: $16 + 18 \div 6 - 3 \times 3$ a) 13 b) 9 c) 10 d) 12 **Explanation:**

Using BODMAS, division and multiplication are done first.

$$16 + 18 \div 6 - 3 \times 3$$

$$= 16 + 3 - 9$$

$$= 19 - 9$$

$$= 10$$

Answer: C) 10

Q. 95

Find slope of line $9x + 5y = 8$.

a) $-\frac{9}{5}$ ✓

b) $\frac{9}{5}$ ✗

c) $-\frac{5}{9}$ ✗

d) $\frac{5}{9}$ ✗

Explanation:

$$9x + 5y = 8$$

Formula:

For a line $ax + by + c = 0$,

Slope (m) = $-\frac{a}{b}$

Here,

$a = 9, b = 5$

Therefore,

$m = -\frac{9}{5}$

Answer = $-\frac{9}{5}$ (Option A)

Q. 96

Find the midpoint of the segment joining $(-8, 4)$ and $(6, -8)$.a) $(-1, -2)$ ✓b) $(-2, -1)$ ✗c) $(1, -2)$ ✗d) $(-1, 2)$ ✗ $A = (-8, 4)$ $B = (6, -8)$

Formula:

Midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$ **Explanation:**

$$= \left(\frac{-8 + 6}{2}, \frac{4 - 8}{2} \right)$$

$$= \left(-\frac{2}{2}, -\frac{4}{2} \right)$$

$$= (-1, -2)$$

Answer = $(-1, -2)$ (Option A)

Q. 97

An 8% discount and a 5% discount in succession is equivalent to a total discount of ____.

- a) 13.4% 
- b) 12.5% 
- c) 13.5% 
- d) 12.6% 

Explanation:

$$\text{Equivalent Discount} = a + b - \left(\frac{ab}{100}\right)$$

$$= 8 + 5 - \frac{8 \times 5}{100}$$

$$= 13 - 0.4$$

$$= 12.6\%$$

Answer: D) 12.6%

Q. 98

If the cubic equation $5a^3 - 3a^2 + 5a - 2 = 0$ has the roots x, y, z , then find the value of $2(x + y + z)$.

a) $3/5$ b) $6/5$ c) $7/5$ d) $2/5$ **Explanation:**

Given cubic equation:

$$5a^3 - 3a^2 + 5a - 2 = 0$$

For a cubic equation:

$$Ax^3 + Bx^2 + Cx + D = 0$$

Formula:

$$\text{Sum of roots} = -\frac{B}{A}$$

Therefore,

$$x + y + z = -\left(\frac{-3}{5}\right) = \frac{3}{5}$$

Now,

$$2(x + y + z) = 2 \times \frac{3}{5} = \frac{6}{5}$$

Answer = $6/5$ (Option B)

Q. 99

Neha, Rajat and Avinash invested ₹1,210, ₹1,650 and ₹1,300, respectively to start a business. If the profit at the end of the year is ₹1,200, what is Avinash's share of the profit?

- a) ₹377 
- b) ₹374 
- c) ₹375 
- d) ₹376 

Explanation:

Since the investment period is the same,

Profit-sharing ratio = Investment ratio

Neha : Rajat : Avinash
= 1210 : 1650 : 1300
= 121 : 165 : 130

Total ratio = 121 + 165 + 130 = 416

Avinash's share

$$\begin{aligned} &= 1200 \times \frac{130}{416} = \text{₹}375 \\ &= 1200 \times \frac{5}{16} \end{aligned}$$

Answer: C) ₹375

Q. 100

If a train runs at 40 km/hr, it reaches its destination late by 10 min, but if it runs at 45 km/hr, it reaches late by 4 min. Find the correct time for the train to complete the journey.

- a) 34 min 
- b) 45 min 
- c) 44 min 
- d) 24 min 

Difference in late time = $10 - 4 = 6 \text{ min} = 1/10 \text{ hour}$

Formula:

$$\text{Distance} = \frac{S_1 \times S_2}{S_2 - S_1} \times \text{Difference in time}$$

$$= \frac{40 \times 45}{45 - 40} \times \frac{1}{10}$$

$$= \frac{1800}{5} \times \frac{1}{10}$$

$$= 36 \text{ km}$$

Time taken at 45 km/hr:

$$= 36/45 \text{ hour} = 48 \text{ min}$$

But the train is still 4 min late.

$$\text{Correct time} = 48 - 4 = 44 \text{ min}$$

Answer: c) 44 min

Explanation: